1.

(a) $J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log(h(x^{(i)})) + (1-y^{(i)}) \log(1-h(x^{(i)}))$

$$\nabla_\theta J(\theta) = \begin{bmatrix} -\frac{1}{m}\sum_{i=1}^{m}(y^{(i)} - h(x^{(i)}))\, x_1^{(i)} \\ \vdots \\ -\frac{1}{m}\sum_{i=1}^{m}(y^{(i)} - h(x^{(i)}))\, x_n^{(i)} \end{bmatrix}$$

$H = \nabla_\theta^2 J(\theta)$, $H_{ij} = \frac{1}{m}\sum_{i=1}^{m} h(x^{(i)})(1-h(x^{(i)}))\, x_i x_j$

$$H = \frac{1}{m} X^T \operatorname{diag}\left(h(x^{(1)})(1-h(x^{(1)})), \cdots, h(x^{(m)})(1-h(x^{(m)})) \right) X$$

$$= \frac{1}{m} X^T W X$$

H is symmetric

$$z^T H z = \frac{1}{m} z^T X^T W X z = \frac{1}{m} (Xz)^T W (Xz) = \frac{1}{m} y^T W y$$

$$= \frac{1}{m}\sum_{i=1}^{m} (y_i)^2 W_{ii} \geq 0. \quad (\because W_{ii} \geq 0)$$

(c)

$$p(y=1|x) = \frac{p(x|y=1)p(y=1)}{p(x|y=1)p(y=1) + p(x|y=0)p(y=0)}$$

$$= \frac{\phi \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right)}{\phi \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right) + (1-\phi)\exp\left(-\frac{1}{2}(x-\mu_0)^T\Sigma^{-1}(x-\mu_0)\right)}$$

$$= \frac{1}{1 + \left(\frac{1}{\phi}-1\right)\exp\left(-\frac{1}{2}(x-\mu_0)^T\Sigma^{-1}(x-\mu_0) + \frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right)}$$

$$(x-\mu_0)^T\Sigma^{-1}(x-\mu_0) - (x-\mu_1)^T\Sigma^{-1}(x-\mu_1)$$

$$= (x^T\Sigma^{-1}x - 2\mu_0^T\Sigma^{-1}x + \mu_0^T\Sigma^{-1}\mu_0) - (x^T\Sigma^{-1}x - 2\mu_1^T\Sigma^{-1}x + \mu_1^T\Sigma^{-1}\mu_1)$$

$$= 2(\mu_1^T\Sigma^{-1} - \mu_0^T\Sigma^{-1})x + \mu_0^T\Sigma^{-1}\mu_0 - \mu_1^T\Sigma^{-1}\mu_1$$

$$\therefore p(y=1|x) = \frac{1}{1 + \exp\left((\mu_0-\mu_1)^T\Sigma^{-1}x + \frac{1}{2}(\mu_1^T\Sigma^{-1}\mu_1 - \mu_0^T\Sigma^{-1}\mu_0) + \ln\left(\frac{1}{\phi}-1\right)\right)}$$

$$\therefore \theta = \Sigma^{-1}(\mu_1-\mu_0)$$

$$\theta_0 = \frac{1}{2}(\mu_0^T\Sigma^{-1}\mu_0 - \mu_1^T\Sigma^{-1}\mu_1) + \ln\left(\frac{1}{1-\phi}-1\right)$$

(d)

$$l(\phi, \mu_0, \mu_1, \Sigma) = \log \prod_{i=1} p(x^{(i)}, y^{(i)})$$

$$= \log \prod_{i=1}^{m} p(x^{(i)}|y^{(i)})\, p(y^{(i)})$$

$$= \log \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi|\Sigma|}} \exp\left(-\frac{(x-\mu_{y^{(i)}})^2}{2\Sigma}\right) \phi^{y^{(i)}}(1-\phi)^{1-y^{(i)}}$$

$$\phi: \quad C \cdot \log \prod_{i=1}^{m} \phi^{y^{(i)}} \cdot (1-\phi)^{1-y^{(i)}}$$

$$= C \cdot \sum_{i=1}^{m} 1\{y^{(i)}=1\} \log \phi + 1\{y^{(i)}=0\} (1-\phi)$$

$$\frac{\partial}{\partial \phi}(\cdots) = C \cdot \left(\frac{\sum 1\{y^{(i)}=1\}}{\phi} - \frac{\sum 1\{y^{(i)}=0\}}{1-\phi} \right) = 0.$$

$$\frac{\sum 1\{y^{(i)}=1\}}{\phi} = \frac{\sum 1\{y^{(i)}=0\}}{1-\phi}$$

$$\frac{1-\phi}{\phi} = \frac{\sum 1\{y^{(i)}=0\}}{\sum 1\{y^{(i)}=1\}}$$

$$\frac{1}{\phi} = 1 + \frac{\sum 1\{y^{(i)}=0\}}{\sum 1\{y^{(i)}=1\}} = \frac{m}{\sum 1\{y^{(i)}=1\}}$$

$$\therefore \phi = \frac{1}{m} \sum_{i=1}^{m} 1\{y^{(i)}=1\}$$

μ_0 :

$$\frac{\partial}{\partial \mu_0} \log \prod_{i=1}^{m} \exp\left(-\frac{(x^{(i)}-\mu_{y^{(i)}})^2}{2\Sigma}\right)$$

$$= \frac{\partial}{\partial \mu_0} \sum_{i=1}^{m} \left(-\frac{(x^{(i)}-\mu_{y^{(i)}})^2}{2\Sigma}\right)$$

$$= \frac{1}{\Sigma} \sum_{i=1}^{m} 1\{y^{(i)}=0\}(x^{(i)}-\mu_0) = 0$$

$$\therefore \mu_0 = \frac{\sum_{i=1}^{m} 1\{y^{(i)}=0\} x^{(i)}}{\sum_{i=1}^{m} 1\{y^{(i)}=0\}}$$

μ_1 : 같은 방식.

Σ :

$$\frac{\partial}{\partial \Sigma} \log \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi\Sigma}} \exp\left(-\frac{(x^{(i)}-\mu_{y^{(i)}})^2}{2\Sigma}\right)$$

$$= \frac{\partial}{\partial \Sigma}\left(m \log \frac{1}{\sqrt{2\pi\Sigma}} + \sum_{i=1}^{m}\left(-\frac{(x^{(i)}-\mu_{y^{(i)}})^2}{2\Sigma}\right)\right)$$

$$= -\frac{m}{2\Sigma} + \sum_{i=1}^{m} \frac{(x^{(i)}-\mu_{y^{(i)}})^2}{2\Sigma^2} = 0$$

$$\therefore \Sigma = \frac{1}{m} \sum_{i=1}^{m} (x^{(i)}-\mu_{y^{(i)}})^2$$

2.

(a)

$$p(y^{(i)}=1 \mid x^{(i)}) = p(y^{(i)}=1 \mid t^{(i)}=1, x^{(i)})\, p(t^{(i)}=1 \mid x^{(i)}) + p(y^{(i)}=1 \mid t^{(i)}=0, x^{(i)})\, p(t^{(i)}=0, x^{(i)})$$

$$= p(y^{(i)}=1 \mid t^{(i)}=1)\, p(t^{(i)}=1 \mid x^{(i)}) + \underbrace{p(y^{(i)}=1 \mid t^{(i)}=0)}_{=0}\, p(t^{(i)}=0 \mid x^{(i)})$$

$$= p(y^{(i)}=1 \mid t^{(i)}=1)\, p(t^{(i)}=1 \mid x^{(i)})$$

$p(y^{(i)}=1 \mid t^{(i)}=1)$, 즉 true 라벨이 라벨링 되어있을 비율이 α로 정해져 있다면.

$$p(y^{(i)}=1 \mid x^{(i)}) = \alpha \cdot p(t^{(i)}=1 \mid x^{(i)})$$

$$\therefore p(t^{(i)}=1 \mid x^{(i)}) = p(y^{(i)}=1 \mid x^{(i)}) / \alpha$$

(b)

$$h(x^{(i)}) \approx p(y^{(i)}=1 \mid x^{(i)})$$

$$= \alpha \cdot p(t^{(i)}=1 \mid x^{(i)})$$

$$\approx \alpha$$

(a)

$$p(y;\lambda) = \frac{e^{-\lambda}\cdot\lambda^{y}}{y!} = \frac{1}{y!}\exp(y\log\lambda - \lambda)$$

$$b(y) = \frac{1}{y!}$$

$$T(y) = y$$

$$\eta = \log\lambda, \quad \lambda = e^{\eta}$$

$$a(\eta) = e^{\eta}$$

(b) $g(\eta) = E[y; \eta] = \frac{\partial}{\partial \eta} a(\eta) = e^{\eta}$

(c)

$$l(\theta) = \log \prod_{i=1}^{m} p(y|x;\theta)$$

$$= \log \prod_{i=1}^{m} \frac{e^{-\lambda} \cdot \lambda^{y}}{y^{(i)}!}, \quad \left(\eta = \theta^T x, \quad h_\theta(x) = \lambda = e^{\eta} = e^{\theta^T x} \right)$$

$$= \log \prod_{i=1}^{m} \frac{1}{y^{(i)}} \exp\left(y^{(i)} \log h_\theta(x^{(i)}) - h_\theta(x^{(i)}) \right)$$

$$= \sum_{i=1}^{m} \log \frac{1}{y^{(i)}} + \sum_{i=1}^{m} \left(y^{(i)} \log h_\theta(x^{(i)}) - h_\theta(x^{(i)}) \right)$$

$$\frac{\partial}{\partial \theta_j} l(\theta) = \sum_{i=1}^{m} \left(y^{(i)} x_j^{(i)} - h_\theta(x^{(i)}) x_j^{(i)} \right)$$

$$= \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right) x_j^{(i)}$$

$$\theta_j := \theta_j + \alpha \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right) x_j^{(i)}$$

If $l(\theta) = \log p(y|x;\theta) \rightarrow$ stochastic g.a

$$\therefore \theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)}) \right) x_j^{(i)}$$

4.

(a) $p(y;\eta) = b(y)\exp(\eta y - a(\eta))$

$$\frac{\partial}{\partial\eta}\int p(y;\eta)\,dy = \int \frac{\partial}{\partial\eta} p(y;\eta)\,dy$$

$$= \int \frac{\partial}{\partial\eta} b(y)\exp(\eta y - a(\eta))\,dy$$

$$= \int b(y)\left(y - \frac{\partial}{\partial\eta}a(\eta)\right)\exp(\eta y - a(\eta))\,dy$$

$$= \int y\cdot b(y)\exp(\eta y - a(\eta))\,dy - \frac{\partial}{\partial\eta}a(\eta)\int b(y)\exp(\eta y - a(\eta))\,dy$$

$$= E[y;\eta] - \frac{\partial}{\partial\eta}a(\eta)$$

$$\left(\int p(y;\eta)\,dy = 1\right)$$

$$= E[y;\eta] - \frac{\partial}{\partial\eta}a(\eta)$$

$\int p(y;\eta)\,dy = 1$, $\quad \frac{\partial}{\partial\eta}\int p(y;\eta)\,dy = 0$.

$\therefore E[y;\eta] - \frac{\partial}{\partial\eta}(a(\eta)) = 0$.

$\therefore E[y;\eta] = \frac{\partial}{\partial\eta}a(\eta)$

(b) $$\frac{\partial^2}{\partial\eta^2}\int p(y;\eta)\,dy = \int b(y)\left\{(y - \frac{\partial}{\partial\eta}(a(\eta))^2\exp(\eta y - a(\eta)) - \frac{\partial^2}{\partial\eta^2}a(\eta)\exp(\eta y - a(\eta))\right\}dy$$

$$= \int b(y)(y-\frac{\partial}{\partial\eta}a(\eta))^2\exp(\eta y - a(\eta))\,dy - \int b(y)\frac{\partial^2}{\partial\eta^2}a(\eta)\exp(\eta y - a(\eta))\,dy$$

$$= E[(y-\mu)^2] - \frac{\partial^2}{\partial\eta^2}a(\eta)\int b(y)\exp(\eta y - a(\eta))\,dy$$

$$= \mathrm{Var}[y;\eta] - \frac{\partial^2}{\partial\eta^2}a(\eta) = 0.$$

$\therefore \mathrm{Var}[y;\eta] = \frac{\partial^2}{\partial\eta^2}a(\eta)$

(c)

$$\ell(\theta) = \log \prod_{i=1}^{m} p(y|x;\theta)$$

$$= \log \prod_{i=1}^{m} b(y^{(i)}) \exp(\theta^T x^{(i)} y^{(i)} - a(\theta^T x^{(i)}))$$

$$= \sum_{i=1}^{m} \log b(y^{(i)}) + \sum_{i=1}^{m} (\theta^T x^{(i)} y^{(i)} - a(\theta^T x^{(i)}))$$

$$\text{NLL}: \; -\ell(\theta) = \sum_{i=1}^{m} (a(\theta^T x^{(i)}) - \theta^T x^{(i)} y^{(i)} - \log b(y^{(i)}))$$

$$\nabla_\theta -\ell(\theta) = \begin{bmatrix} \sum_{i=1}^{m} a'(\theta^T x^{(i)}) x_1^{(i)} - x_1^{(i)} y^{(i)} \\ \vdots \\ \sum_{i=1}^{m} a'(\theta^T x^{(i)}) x_n^{(i)} - x_n^{(i)} y^{(i)} \end{bmatrix}$$

$$\nabla_\theta^2 -\ell(\theta) = \begin{bmatrix} \sum_{i=1}^{m} a''(\theta^T x^{(i)}) x_1^{(i)} x_1^{(i)} & \cdots & \sum a''(\,) x_1^{(i)} x_n^{(i)} \\ \vdots & \ddots & \vdots \\ \sum a''(\,) x_n^{(i)} x_1^{(i)} & \cdots & \sum a''(\,) x_n^{(i)} x_n^{(i)} \end{bmatrix}$$

$$= X^T W X$$

$$\left(W = \text{diag}(a''(\theta^T x^{(1)}), a''(\theta^T x^{(2)}), \cdots, a''(\theta^T x^{(m)})) \right)$$

$z^T H z$ (for any z)

$$= z^T X^T W X z = (Xz)^T W (Xz)$$

$$= y^T W y \quad \text{for any } y$$

$$= \sum_{i=1}^{m} y_i^2 a''(\theta^T x^{(i)}) \geq 0$$

$$(\because a''(\theta^T x^{(i)}) = \text{Var}[Y|X;\theta] \geq 0.)$$

$\therefore H \succeq 0$ (PSD)

5.

(a)

i. $W = \text{diag}\left(\frac{1}{2}w^{(1)}, \frac{1}{2}w^{(2)}, \cdots, \frac{1}{2}w^{(m)}\right)$

ii. $\nabla_\theta J(\theta) = \nabla (\theta^T x^T - y^T) W (X\theta - y)$

$= \nabla (\theta^T X^T W X \theta - \theta^T X^T W y - y^T W X \theta + y^T y)$

$= \nabla_\theta (X\theta)^T W X \theta - 2y^T W X \theta + y^T y)$

$= 2X^T W X \theta - 2X^T W y = 0.$

$\theta = (X^T W X)^{-1} X^T W y$

(b)

$$l(\theta) = \log \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma^{(i)}} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}\right)$$

$$= \sum_{i=1}^{m} \log \frac{1}{\sqrt{2\pi}\sigma^{(i)}} - \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}$$

maximize $l(\theta)$

$$= \text{minimize} \sum \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}$$

$$= \text{minimize} \quad \frac{1}{2} \sum_{i=1}^{m} \frac{1}{(\sigma^{(i)})^2} \cdot (\theta^T x^{(i)} - y^{(i)})^2 = J(\theta).$$

$$w^{(i)} = \frac{1}{(\sigma^{(i)})^2}$$

$$\frac{\partial}{\partial \theta_j} l(\theta) = \sum \frac{1}{(\sigma^{(i)})^2} (y^{(i)} - \theta^T x^{(i)}) x_j^{(i)}$$